

C5.1 How are chemicals separated and tested for purity?

Teaching and learning narrative	Assessable learning outcomes <i>Learners will be required to:</i>	Linked learning opportunities
Many useful products contain mixtures. It is important that consumer products such as drugs or personal care products do not include impurities. Mixtures in many consumer products contain pure substances mixed together in definite proportions called formulations. Pure substances contain a single element or compound. Chemists test substances made in the laboratory and in manufacturing processes to check that they are pure. One way of assessing the purity of a substance is by testing its melting point; pure substances have sharp melting points and can be identified by matching melting point data to reference values. Chromatography is used to see if a substance is pure or to identify the substances in a mixture. Components of a mixture are identified by the relative distance travelled compared to the distance travelled by the solvent. Rf values can be calculated and used to identify unknown components by comparison to reference samples. Some substances are insoluble in water, so other solvents are used. Chromatography can be used on colourless substances but locating agents are needed to show the spots. Preparation of chemicals often produces impure products or a mixture of products. Separation processes in both the laboratory and in industry enable useful products to be separated from by-products and waste products. The components of mixtures are separated using processes that exploit the different properties of the components, for example state, boiling points or solubility in different solvents. Separation processes are rarely completely successful and mixtures often need to go through several stages or through repeated	1. explain that many useful materials are formulations of mixtures	Specification links: - particle model and changes of state (C1.1) - Fractional distillation of crude oil on an industrial scale (C3.4) Ideas about Science - use the particle model to explain the idea of a pure substance
	2. explain what is meant by the purity of a substance, distinguishing between the scientific and everyday use of the term 'pure'	
	3. use melting point data to distinguish pure from impure substances	
	4. recall that chromatography involves a stationary and a mobile phase and that separation depends on the distribution between the phases	
	5. interpret chromatograms, including calculating Rf values M3c	
	6. suggest chromatographic methods for distinguishing pure from impure substances <i>PAG4</i> Including the use of: a) paper chromatography b) aqueous and non-aqueous solvents c) locating agents	
	7. describe, explain and exemplify the processes of filtration, crystallisation, simple distillation, and fractional distillation <i>PAG3, PAG7</i>	
	8. suggest suitable purification techniques given information about the substances involved <i>PAG3, PAG7</i>	
